

JADE

BUILD MANUAL · REVISION 2 · AUGUST 2026

Everything you need to run, extend and ship Jade — corrected against what actually happened.

Revision 1 was written before any of it had been built. This one is written after, and every command in it has been run.

What changed since revision 1 — the terminal does not treat `#` as a comment the way that manual claimed; the app runs on port 3100, not 3000; Vercel needs *no* config file and one specific Root Directory; Neon's default role silently bypasses the security that isolates client data; and Next reads environment files from a directory nobody expects. Each of those cost real time. Each is written down here.

Companion to the Jade master plan and the `jadev2` repository.

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01

Where Jade actually is

Read this before anything else. Knowing precisely what exists stops you building on something that does not.

PHASE	STATE	WHAT THAT MEANS
0 · Rails	Done	Monorepo, CI, Sentry scrubber that redacts birth data, v0 prototype preserved at <code>/legacy</code>
1 · Calculation core	Partial	Sidereal positions including Rāhu and Ketu, all sixteen vargas, whole-sign houses, nakṣatras, Vimśottarī to five levels, pañcāṅga, dignity, combustion, dṛṣṭi
2 · People	Done	Database with forced row-level security, place search, historical timezones, add-a-person, content-addressed chart cache
3 · Chart workspace	Partial	North and South Indian charts, ṣoḍaśavarga contact sheet, Vimśottarī column, pañcāṅga card
7 · Launch	Partial	Live at jadeapp.co with emailed sign-in. No billing yet

What is deliberately not built

Four things were attempted or specified and then left out on purpose. None is an oversight, and each has a written test that would let it ship.

MISSING	WHY IT WAS NOT GUESSED
Aṣṭakavarga	Fifty-six rows of benefic-point tables. Recalled approximately, they produce a chart that totals correctly and is wrong row by row. The seven per-graha totals (48, 49, 39, 54, 56, 52, 39, summing to 337) are the check that would prove them.
Ṣaḍbala	Six sources with sub-components, and authorities differ on several. Needs a per-component convention agreed and recorded first.
Yogas	A yoga list without cancellation rules is astrologically dishonest, and cancellations are where schools diverge most. Pañca Mahāpuruṣa is unambiguous and would be the honest starting point.
East Indian chart	Built, rendered, and cut. The Bengali sign arrangement could not be verified against a reference, and a plausible guess at a regional convention is exactly what a Bengali astrologer spots in one glance.

THE STANDARD BEING HELD TO

Everything that ships is checked against Swiss Ephemeris, which is what professional astrologers compare against. 464 tests, of which 408 are in the calculation core alone, plus three that drive a real browser. Anything that cannot be checked does not ship. That rule is the product.

macOS. Windows and Linux differ only where noted.

REVISION 1 GOT THIS WRONG – READ IT BEFORE YOU PASTE ANYTHING

The old manual said lines beginning with `#` are comments you can safely paste. **In zsh, which is what macOS uses, that is false by default.** Pasting `pnpm test # expect 117 passed` sends `#`, `expect`, `117` and `passed` to the command as arguments, and you get an error that looks like a broken project.

Fix it once and the grey notes in this manual become safe:

```
echo 'setopt interactive_comments' >> ~/.zshrc
source ~/.zshrc
```

Until you do that, paste only the command itself and leave the grey text behind.

Install the tools

```
# Homebrew first. Run the "Next steps" lines it prints when it finishes.
/bin/bash -c "$(curl -fsSL \
  https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

brew install git gh fnm jq flyctl
brew install --cask visual-studio-code
brew install python@3.12
```

Node, via fnm. Jade needs 20 or newer; Claude Code's npm package wants 22, so install 22 and stop thinking about it:

```
echo 'eval "$(fnm env --use-on-cd --shell zsh)"' >> ~/.zshrc
source ~/.zshrc
fnm install 22
fnm default 22
node --version
```

IF THAT PRINTS V25 OR ANYTHING OTHER THAN V22

`fnm default` only affects *new* shells. Run `fnm use 22` in the current one, or open a new tab.

```
corepack enable      turns on pnpm, which ships with Node
pnpm --version
python3 -m pip install --user pyswisseph
curl -fsSL https://claude.ai/install.sh | bash
claude --version
```

If Claude Code installs but `claude` is not found, it will tell you to add `~/local/bin` to your path. Do what it says, then `source ~/.zshrc`.

Tell Git who you are

```
git config --global user.name "Nalu Gramberg"  
git config --global user.email "ngramberg808@gmail.com"  
git config --global init.defaultBranch main  
git config --global pull.rebase true  
gh auth login
```

Choose GitHub.com → SSH → generate a new key → log in with a browser. Confirm with `ssh -T git@github.com`.

Running it locally

1 Get the code.

```
cd ~/Projects/jade
git pull
```

2 Install. First run takes a minute; later runs take seconds.

```
pnpm install
```

3 Run the tests. This proves the calculation core agrees with Swiss Ephemeris on your machine, not just on someone else's.

```
pnpm test
```

Expect 445 passing without a database. The database tests skip unless you point `TEST_DATABASE_URL` at a scratch database — never at Neon, because they truncate every table. A guard refuses if the URL does not look disposable.

4 Start it.

```
pnpm --filter @jade/web dev
```

Open **`http://localhost:3100`**. Not 3000 — that port collides with another project of yours whose service worker intercepts requests, which makes Jade look broken when it is running perfectly.

WHERE THE ENVIRONMENT FILE GOES, AND WHY IT IS NOT OBVIOUS

`.env.local` lives at the **repository root**, next to `package.json` — not inside `apps/web`.

Next.js normally reads env files from the Next project directory, which here is `apps/web`. A single file at the root would be invisible to it. Jade's `next.config.mjs` loads the root file explicitly before Next boots, so one file serves the whole monorepo. Real environment variables still win, which is what Vercel needs.

Next reads env files **once, at boot**. Change one and you must restart the dev server.

If the page says *"Jade just needs a database"*, that is the app telling you `DATABASE_URL` did not reach it. Check the file is at the root and restart.

04 The database

Postgres on Neon. Two connection strings, two roles, and one trap that silently disables the security protecting client data.

Two URLs, and they are not interchangeable

VARIABLE	ENDPOINT	USED BY
<code>DATABASE_URL</code>	the one with -pooler in the hostname	the app, constantly opening and dropping connections
<code>DIRECT_DATABASE_URL</code>	the one without	migrations, which take advisory locks a transaction pooler cannot hold

Strip `channel_binding=require` from both. It is a libpq parameter; the driver Jade uses does not implement SCRAM channel binding and it can surface as a connect-time error. `sslmode=require` still encrypts the connection.

```
pnpm db:migrate    schema, row-level security, trigram index
pnpm db:seed       20 starter cities
pnpm db:doctor     prove the isolation actually holds
```

NEON'S DEFAULT ROLE BYPASSES ROW-LEVEL SECURITY

`neondb_owner` is a member of `neon_superuser`, which carries the `BYPASSRLS` attribute. A role with that attribute skips every security policy unconditionally. Connect the app as it and workspaces are **not isolated at all** — the policies exist, the tests pass, and production leaks. Supabase's `postgres` role is the same story.

One command fixes it. It creates a second role that owns nothing and can bypass nothing, and rewrites `DATABASE_URL` itself so the new password never passes through a clipboard:

```
pnpm db:app-role -- --write
restart the dev server, then:
pnpm db:doctor
```

The owner keeps running migrations through `DIRECT_DATABASE_URL`. Only the app's connection changes.

What db:doctor checks

It does not take configuration's word for anything. It reads the live role's privileges, confirms row-level security is enabled *and forced* on every workspace-scoped table, then creates a throwaway row and tries to read it from a different workspace and from an unscoped connection. If either succeeds, it fails loudly and tells you not to put real client data in that database.

A DETAIL WORTH KNOWING

`BYPASSRLS` is **not inherited through role membership** — verified on Postgres 16. A member of a bypassing role still has its inserts refused and its reads filtered. What matters is whether the attribute sits directly on the role you connect as, which is exactly what the doctor reads. Neon sets it directly, which is why the fix is needed.

The full place atlas

The seed loads twenty cities so the app works immediately. For real use:

```
pnpm add -w -D all-the-cities geo-tz
pnpm places:import
```

That loads about 135,000 cities with time zones resolved from boundary polygons rather than guessed from the country — which is why Ann Arbor comes out as America/Detroit and not America/New_York.

Vercel, from the `main` branch. There is no config file in the repository, and that is deliberate.

THE DOUBLED PATH THAT READS LIKE A BUILD FAILURE

If you see this, the build *succeeded* and only the output collection failed:

The Next.js output directory "apps/web/.next" was not found at
"/vercel/path0/apps/web/apps/web/.next"

`apps/web` twice. Root Directory is already `apps/web`, and an Output Directory setting of `apps/web/.next` resolves relative to it. Clear the override.

There is a second layer: when a Root Directory is set, Vercel reads `vercel.json` from *inside that directory*, not from the repository root. A root-level config file sits there looking authoritative and is never read. Jade has none for that reason.

Build settings

SETTING	VALUE
Root Directory	<code>apps/web</code>
Framework Preset	Next.js — auto-detected
Build Command	default , no override
Output Directory	default , no override
Install Command	default , no override

Vercel detects `pnpm-workspace.yaml` and installs from the repository root, so the shared packages are available even though the root directory is `apps/web`. That matters because the web app imports them as TypeScript source, not as built artifacts.

Environment variables

Set every one for Production, Preview *and* Development.

VARIABLE	VALUE
<code>AUTH_MODE</code>	<code>supabase</code> — dev mode throws in production, on purpose
<code>NEXT_PUBLIC_APP_URL</code>	<code>https://jadeapp.co</code>
<code>DATABASE_URL</code>	the restricted role, pooled endpoint
<code>DIRECT_DATABASE_URL</code>	the owner role, direct endpoint
<code>NEXT_PUBLIC_SUPABASE_URL</code>	Supabase → Settings → API
<code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code>	Supabase → Settings → API

VERCEL WILL WARN ABOUT THE NEXT_PUBLIC_ PREFIX ON THE SUPABASE VALUES

That warning is right in general and wrong here. Both of those are designed to be public — the key is literally prefixed `sb_publishable_` and ships in every browser bundle of every Supabase app. They *must* carry the prefix, because the sign-in form runs in the browser. The key that must never carry it is the Supabase **service role** key, which Jade does not use. Switch the two to "Config" if you want the triangle gone.

TWO CREDENTIALS TO ROTATE BEFORE ANYONE ELSE USES JADE

While we were setting this up you pasted your **Neon database password** and your **Resend API key** into a chat window. Nothing bad happened, but a secret that has been through a transcript is no longer a secret, and both of those still work right now.

Rotate them, in this order:

1. **Neon** → your project → Roles → `neondb_owner` → Reset password. Copy the new connection strings into `.env.local` and into Vercel, then run `pnpm db:app-role -- --write` again so the app's restricted role gets a fresh password too.
2. **Resend** → API Keys → revoke the old key, create a new one, update it wherever it is set. It is not wired into Jade yet, so nothing breaks while it is missing.
3. Redeploy, then confirm with `pnpm db:doctor`.

The general rule from here: secrets go into `.env.local` and Vercel and nowhere else. Never into a chat, a commit, a screenshot, or a note.

Deploying is one command

```
git push
```

RUN THIS BEFORE YOU PUSH, NOT AFTER VERCEL TELLS YOU

Two classes of failure exist only in a production build and never in the dev server: type errors that `next build` checks and `next dev` does not, and a `pnpm-lock.yaml` that has drifted from a `package.json`. Vercel installs with `--frozen-lockfile`, so the second one stops the build before it starts.

```
pnpm install --frozen-lockfile
pnpm build
```

Green here means green on Vercel. It takes about a minute and saves the round trip.

Vercel builds on push. If the site shows an older version, check which branch it is building: Vercel deploys `main`, and work sitting on a feature branch will never appear no matter how many times you push it.

Emailed links, no passwords. For a product holding other people's birth data, not storing passwords is the right default.

Two modes

AUTH_MODE	BEHAVIOUR
dev	Sign in with any email, no password, no external service. The whole app runs on a database alone. Throws if it ever starts in production.
supabase	Real sign-in. Supabase emails a one-time link; a middleware refreshes the session on every request.

MAGIC LINKS FAIL SILENTLY WITHOUT THIS

Supabase → Authentication → URL Configuration:

- **Site URL:** `https://jadeapp.co`
- **Redirect URLs:** `https://jadeapp.co/auth/callback`, `https://*.vercel.app/auth/callback`, `http://localhost:3100/auth/callback`

The Site URL alone is not enough. Without the callback path in the allow-list, the link arrives, you click it, and you land back on the sign-in page with no useful explanation.

Supabase's built-in sender is rate-limited to a handful of messages an hour on the free tier. Fine for two people. When it starts to pinch, plug your Resend key in as custom SMTP under Authentication → Emails.

The map of the repository

PATH	WHAT LIVES THERE
<code>CLAUDE.md</code>	The project constitution. Claude Code reads it before every task.
<code>docs/</code>	Vision, architecture, domain model, the calculation spec, phases, launch, accuracy
<code>packages/astro/</code>	The calculation core. Pure mathematics — no database, no network, no clock
<code>packages/astro/test/</code>	Golden fixtures generated from Swiss Ephemeris, plus the invariant suite
<code>packages/ui/</code>	Design tokens and the chart components
<code>packages/db/</code>	Schema, migrations, row-level security, queries, and the doctor
<code>packages/atlas/</code>	Place search and historical timezone resolution
<code>packages/interpret/</code>	Grounded interpretation — Phase 8, still a stub
<code>apps/web/</code>	The Next.js app
<code>apps/worker/</code>	Long-running jobs — Phase 5, still a health endpoint
<code>legacy/</code>	The original single-file prototype, served at <code>/legacy</code>
<code>_to_delete/</code>	Scratch. Where files get parked because the sandbox writing to your machine cannot delete. Safe to <code>rm -rf</code> any time.

THE ONE ARCHITECTURAL RULE

`packages/astro` must never import the database, the web app, or anything that touches the network or reads the clock. Lint enforces it and CI fails if it is broken. That purity is what makes charts cacheable, testable, portable to a phone, and one day sellable as an API. Everything else is replaceable; this is the asset.

Working with Claude Code

- 1 **Branch.** `git checkout -b phase-4-relationships`
- 2 **Open Claude Code in the repo.** `cd ~/Projects/jade && claude`
- 3 **Paste the phase prompt** from `docs/05-phases.md`, verbatim.
- 4 **Read what it proposes before it writes.** Ask for types first on anything substantial.
- 5 **Make it run the tests.** Not done until `pnpm test`, `typecheck` and `lint` are green.
- 6 **Review the diff yourself.** `git diff main`
- 7 **Commit, push, merge to main.** Vercel deploys from `main`.

Reviewing what you cannot fully read

You will not verify a *ṣaḍbala* implementation line by line, and you do not need to. Check these five things instead:

- **Does it cite a source?** Every exported function in the core should name the classical text. Uncited astrology is invented astrology.
- **Does the test compare against something external?** Swiss Ephemeris, Jagannātha Hora, a printed worked example — not its own output.
- **Did a tolerance get loosened?** A change in `test/accuracy.test.ts` is a red flag unless the commit explains exactly why.
- **Are disagreements surfaced or buried?** Where schools differ, the code should offer a named option and add a line to the spec.
- **Is the purity rule intact?** No database or network imports inside `packages/astro`, and lint not weakened to allow one.

THE MOST USEFUL SENTENCE YOU CAN TYPE

"Before you write anything, tell me what you are about to change and why." One exchange, and it prevents most of the rework.

AND THE MOST USEFUL THING TO INSIST ON

Ask for it to be *run*, not just compiled. Every serious bug in this project so far — a submit button that swallowed its own form, a cache key that collided across tenants, grahas rendering in a shuffled order, sign numbers landing on top of glyphs — passed typecheck, lint and unit tests, and appeared the moment a browser loaded the page.

The accuracy program

The part competitors cannot copy quickly, and the reason a professional would switch.

Jade's test fixtures are not hand-written. A Python script drives Swiss Ephemeris — the reference every professional astrologer compares against — and writes out eighteen complete charts. The TypeScript suite then computes the same charts and asserts agreement within stated tolerances. Diverge and CI goes red.

```
python3 -m pip install --user pyswisseph
python3 scripts/generate_fixtures.py          regenerate
python3 scripts/generate_fixtures.py --check  CI drift gate
WRITE_ACCURACY_REPORT=1 pnpm --filter @jade/astro test
```

What the eighteen charts cover

Each exists to catch a class of bug, and the reason is stored beside the data: a pre-standard-time 1881 London birth, an 1847 chart that stretches ΔT , a birth above the Arctic Circle, southern hemisphere, equatorial, a leap day, exact midnight and noon, an eclipse day, Mercury and Mars retrograde, a Saturn station where speed crosses zero, and a 2099 chart where every model is extrapolating.

Worst measured error, current core

QUANTITY	ERROR
Ayanāṁśa, nutation	0.005"
Mean node (Rāhu / Ketu)	0.43"
Sun	1.1"
Ascendant · Midheaven	1.6" · 2.0"
Moon	4.4"
Sunrise / sunset	under 2 min
Saturn · Uranus · Pluto	11" – 13.5"
Neptune	17.4"

Those last two rows are why the CHF 700 Swiss Ephemeris licence matters. The MIT [astronomy-engine](#) is arcsecond-class for the Sun, Moon, nodes and ascendant, and ten to twenty arcseconds out on the outer planets. Fine for a live time-scrubber; an astrologer comparing your Saturn against Jagannātha Hora will see it in the seconds column and conclude the software is wrong.

Three things it will catch that nothing else did

- The ascendant came out **180° wrong above the Arctic Circle** — the standard formula resolves the angle but not which horizon point is rising. Seventeen charts passed; Tromsø failed by exactly half a circle.
- **Trimśāṁśa only ever yields ten signs**. An invariant test asserting all twelve failed — correctly, because the luminaries own no trimśāṁśa. The test was wrong and the code was right.

- **Vāra starts at sunrise.** A birth before dawn belongs to the previous weekday, and the pañcāṅga card says so on screen.

Command reference

Daily

<code>pnpm install</code>	After every pull
<code>pnpm --filter @jade/web dev</code>	The app at localhost:3100
<code>pnpm test</code>	All unit tests
<code>pnpm --filter @jade/astro test</code>	Just the calculation core — fast, run constantly
<code>pnpm typecheck</code> · <code>pnpm lint</code> · <code>pnpm format</code>	Individual CI gates
<code>pnpm build</code>	Full production build
<code>pnpm verify</code>	All of the above, in CI's order. Run this before every push.

Database

<code>pnpm db:migrate</code>	Apply migrations, over the direct endpoint
<code>pnpm db:seed</code>	Twenty starter cities
<code>pnpm db:doctor</code>	Prove workspace isolation holds on this database
<code>pnpm db:app-role -- --write</code>	Create a non-bypassing role and rewrite DATABASE_URL
<code>pnpm places:import</code>	The full 135,000-city atlas

WHY THESE ALL SAY `RUN` INTERNALLY

`pnpm --filter <pkg> doctor` does not work: pnpm intercepts `doctor` as one of its own built-in commands. The scripts use `pnpm --filter <pkg> run doctor` for that reason. If you add a script whose name collides with a pnpm command, this is what you will hit.

Git

<code>git checkout -b name</code>	Start a branch
<code>git add -A && git commit -m "..."</code>	Save a checkpoint
<code>git push</code>	On <code>main</code> , this deploys
<code>gh pr create --fill</code>	Only works once the branch has commits on it
<code>git checkout -- <file></code>	Throw away changes to one file

When something breaks

Every one of these was hit while building Jade. None is hypothetical.

Terminal and setup

WHAT YOU SEE	WHAT IT IS
<code>Could not find task '117' in project</code>	You pasted a command with its grey comment attached and zsh passed it as arguments. Run the <code>set opt interactive_comments</code> line in chapter 02.
<code>node --version</code> prints the wrong major	<code>fnm default</code> only affects new shells. <code>fnm use 22</code> .
<code>zsh: command not found: claude</code>	Installed but not on your path. Add <code>~/local/bin</code> , then <code>source ~/.zshrc</code> .
<code>zsh: command not found: psql</code>	You do not need it. <code>pnpm db:doctor</code> answers the same questions using the driver already in the repo.
<code>Permission denied (publickey)</code>	SSH key not registered. Re-run <code>gh auth login</code> and choose SSH.

Running Jade

"Jade just needs a database" but the URL is set	<code>.env.local</code> is in the wrong place, or the dev server has not been restarted. It belongs at the repository root, and Next reads env files only at boot.
<code>DIRECT_DATABASE_URL is not set</code> from a script	Should not happen any more — the scripts walk up to the workspace root. If it does, the message names the files it searched.
The page loads but shows a different app	Another project's service worker is registered on that port. Jade runs on 3100 specifically to avoid it.
<code>Module not found: Can't resolve './types.js'</code>	The <code>extensionAlias</code> block in <code>apps/web/next.config.mjs</code> was removed. It is what lets webpack resolve the ESM-correct <code>.js</code> specifiers in TypeScript source.
<code>useActionState is not a function</code>	React 19 API on React 18. Next 14 ships React 18; use plain server actions rather than the form hooks.
A form does nothing when clicked	A submit button that disables itself synchronously in its own click handler cancels the submission. Defer the state change.
Port 3100 in use	<code>lsof -ti:3100 xargs kill</code>
<code>__webpack_modules__[moduleId] is not a function</code>	You ran <code>pnpm build</code> while <code>next dev</code> was running. They share <code>.next</code> and the production build overwrites the dev server's chunks out from under it. Stop the dev server, <code>rm -rf apps/web/.next</code> , start it again.

Database

<code>db:doctor</code> reports BYPASSRLS	Expected on a fresh Neon project. <code>pnpm db:app-role -- --write</code> , restart, re-run.
--	---

Two NOTICE blocks on a second <code>db:migrate</code>	Should be silent now. "Already exists, skipping" is success, not failure.
Tests wipe data you wanted	They truncate by design. <code>TEST_DATABASE_URL</code> must point at a scratch database; a guard refuses anything that does not look disposable.
Cache never hits — every page says "computed"	Fixed, but the shape is worth knowing: the cache key must include the workspace, or two practices with identical birth data collide on the primary key while row-level security hides the row from the second.

Deploying

Output directory not found, with a doubled path	Clear the Output Directory override in Vercel. See chapter 05.
Deploy succeeds but the site is an old version	Vercel builds <code>main</code> . Work on a feature branch never appears.
Every page 500s in production	<code>AUTH_MODE</code> is not <code>supabase</code> . Dev mode refuses to serve in production, deliberately.
The email link bounces back to sign-in	The callback path is missing from Supabase's redirect allow-list.
<code>ERR_PNPM_OUTDATED_LOCKFILE</code>	A <code>package.json</code> changed without <code>pnpm-lock.yaml</code> being regenerated. Vercel installs with <code>--frozen-lockfile</code> , so a mismatch that your machine ignores stops the build dead. Run <code>pnpm install</code> , commit the lockfile with the change that caused it.
<code>The inferred type of X cannot be named without a reference to .pnpm/...</code>	A package whose types appear in an exported signature is only reachable transitively. It must be a <i>direct</i> dependency even if nothing imports it by name. This is why <code>@supabase/supabase-js</code> is listed in <code>apps/web</code> . Only <code>next build</code> catches it — never the dev server.

THE UNIVERSAL FIRST MOVE

`rm -rf node_modules .next .turbo && pnpm install`. Two minutes, and it fixes a surprising share of "it worked yesterday".

Before you take money

Birth data is exact time and place of birth — precisely what is used to impersonate someone. Treat it accordingly.

Done already

- Row-level security forced on every workspace-scoped table, and proven by a live probe rather than asserted
- The app connects as a role that cannot bypass those policies
- Sentry scrubs anything resembling birth data before an error leaves the server
- Export and hard delete both work, and export is portable JSON
- No passwords stored — sign-in is an emailed link

Still to do

- ☐ **Column-level encryption of birth data.** Neon encrypts at rest and RLS covers application bugs, but neither stops a leaked database dump. Needs an envelope-encryption design, because encrypting the datetime and coordinates breaks ordering and range queries.
- ☐ Privacy policy naming what is collected and why
- ☐ Terms stating guidance not medical, financial or legal advice — and the no-death, no-disease, no-litigation clause in writing
- ☐ A data processing agreement for professional customers, who hold their clients' data in your product
- ☐ Age gate at 16
- ☐ All secrets rotated once before launch, in case any leaked during development
- ☐ Dependencies scanned — Dependabot on, `pnpm audit` clean
- ☐ Swiss Ephemeris professional licence purchased and the receipt filed
- ☐ GeoNames attribution in the footer and on a `/licenses` page

THE OBLIGATION UNDERNEATH ALL OF IT

A professional who puts their client book into Jade is trusting you with a hundred other people's birth data — people who never agreed to anything with you. That is worth being conspicuously careful about in the marketing as well as the code.

What it costs

ITEM	NOW	FIRST CUSTOMERS	~300 SUBSCRIBERS
Domain	\$12/yr	\$12/yr	\$12/yr
Vercel	free	\$20/mo	\$20–60/mo
Neon	free	free–\$19/mo	\$19–69/mo
Supabase (auth only)	free	free–\$25/mo	\$25/mo
Resend	free	free–\$20/mo	\$20/mo
Sentry	free	free–\$26/mo	\$26/mo
Fly.io (worker, Phase 5)	–	\$5/mo	\$15–40/mo
Stripe	–	2.9% + 30¢ per charge	
Claude Code	\$20–200/mo	\$20–200/mo	\$20–200/mo
Swiss Ephemeris	–	CHF 700 once	–
Monthly	~\$20–200	~\$70–300	~\$150–400

300 Professional subscribers at \$99/month is roughly \$350,000 a year against infrastructure in the low thousands. High margin, if you can find the three hundred people — which is what the free tier, the SEO tools and the published accuracy page exist to do.

Engineering

monorepo	One repository holding several related projects
row-level security	Postgres deciding who may read which rows, so an application bug cannot leak another practice's data
BYPASSRLS	A role attribute that skips every such policy. Managed Postgres providers give it to the default role.
content-addressed	Naming a cached result by a hash of its inputs, so a stale one cannot be served
pure function	Same inputs, same output, no side effects. The whole calculation core.
golden fixture	A saved known-correct result from a trusted reference, that tests compare against
ΔT	The drift between atomic time and the Earth's rotation. Different models disagree for far-past and far-future dates, which is why the Moon's tolerance relaxes outside 1900–2050.

Jyotiṣa

ayanāṁśa	The offset between the tropical and sidereal zodiacs. Schools disagree, so Jade never hides which one it used.
graha	The nine: Sun, Moon, Mars, Mercury, Jupiter, Venus, Saturn, Rāhu, Ketu
Rāhu / Ketu	The lunar nodes. Not physical bodies, and absent from the original prototype entirely.
nakṣatra	One of 27 lunar mansions of 13°20', each divided into four <i>padas</i>
varga	A divisional chart. Jade computes all sixteen, each re-seated on its own ascendant.
vargottama	The same sign in D1 and D9 — a classical mark of strength
daśā	A planetary period. Vimśottarī runs 120 years from the Moon's nakṣatra at birth.
pañcāṅga	The five limbs: tithi, nakṣatra, yoga, karaṇa, vāra
vāra	The weekday — beginning at sunrise, not midnight
dr̥ṣṭi	Vedic aspect. Whole houses, and asymmetric: everything sees the 7th, Mars also the 4th and 8th, Jupiter the 5th and 9th, Saturn the 3rd and 10th.
mūlatrikoṇa	A degree range where a graha is especially strong. The Moon's sits in Venus's Taurus — the one exception.
combustion	A graha too close to the Sun to act. Within one degree it is <i>cazimi</i> , which strengthens rather than burns.
aṣṭakavarga	A point-scoring system for how favourable each sign is. Not yet built — see chapter 01.
whole sign houses	The Vedic default: house one is the entire sign holding the ascendant

NEXT

Phase 4 is relationships — aṣṭakūṭa matching, synastry, the shared daśā timeline. Before that, the honest move is the aṣṭakavarga and yoga work in chapter 01, because those are what make a professional take the tool seriously.